03/12/2025

Code-A

Aakash
Medical|IIT-JEE|Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18, Udyog Vihar, Gurugram, Haryana - 122015, *Ph.*+91-1244168300

MM : 720

NCERT Booster Program for NEET-2026_NBTS-01A

Time : 180 Min.

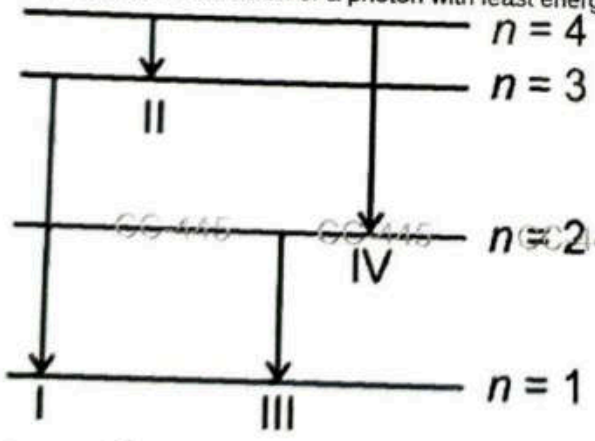
Topics Covered
Physics: Atoms, Nuclei, Semiconductor Electronics, Experimental Skills, Units & Measurements, Motion in a Straight Line
Chemistry: Amines, Biomolecules, The p-Block Elements: (Group 15-18), Principles Related to Practical Chemistry, Some Basic Concepts of Chemistry, Structure of Atom
Botany: Ecosystem, Biodiversity and Conservation, Cell: The Unit of Life, Cell Cycle and Cell Division
Zoology: Biotechnology-Principles and Processes, Biotechnology and Its Applications, Structural Organisation in Animals: Animal Tissues only

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

Vivek Krishna

PHYSICS

1. The diagram shows the energy levels for an electron in a certain atom. Which transition among the following represents the emission of a photon with least energy



(1) II
(2) III
(3) I
(4) IV

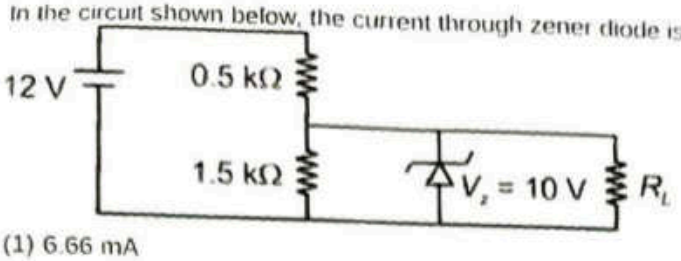
2. A quantity x is defined by $x = \frac{A^2\sqrt{B}}{C^{-2}}$, and the percentage errors in the measurements of A, B and C are 1%, 2% and 0.5% respectively. The maximum percentage error in x is

(1) 2%
(2) 4%
(3) 3%
(4) 6%

3. An intrinsic semiconductor has equal electron and hole concentration of 1.5×10^{16} cm^{-3}. Doping by phosphorus, increases electron concentration to 2.25×10^{22} m^{-3}. The hole concentration in doped semiconductor is

(1) 10^{10} m^{-3}
(2) 10^{9} m^{-3}
(3) 10^{11} m^{-3}
(4) 5×10^{9} m^{-3}

4. In the circuit shown below, the current through zener diode is



(1) 6.66 mA

(2) 4 mA

(3) 2 mA

(4) Zero

5. Given below are two statements:
Statement I: Zero error is always subtracted from measured reading for a vernier calliper.
Statement II: When the jaws are made to touch each other, then zero of main scale and zero of vernier scale always coincide with each other.
In the light of the above statements, choose the most appropriate answer from the options given below.

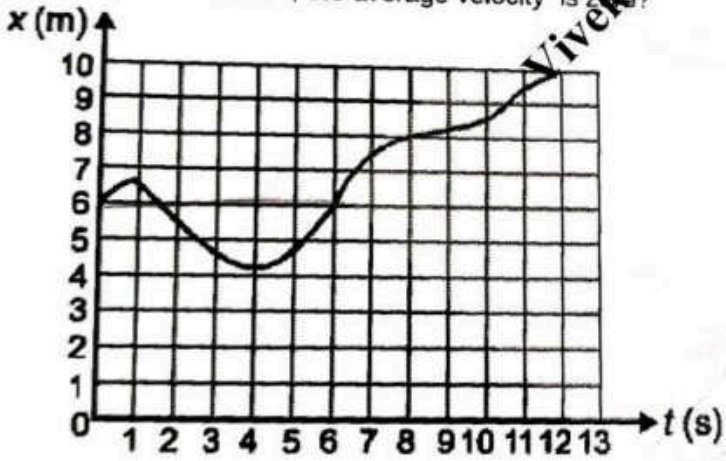
(1) Statement I is correct and statement II is incorrect

(2) Statement I is incorrect and statement II is correct

(3) Both statements are correct

(4) Both statements are incorrect

6. Position-time graph for a particle is shown in figure. Starting from t = 0, at what time t, the average velocity is zero?



(1) 1 s

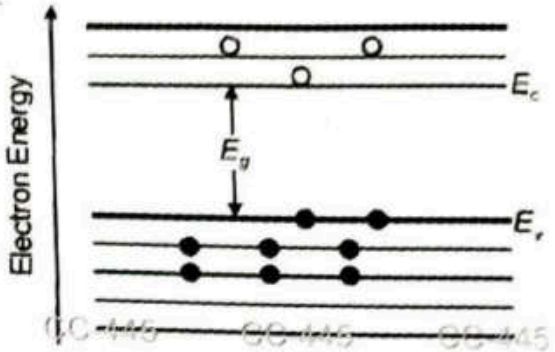
(2) 3 s

(3) 6 s

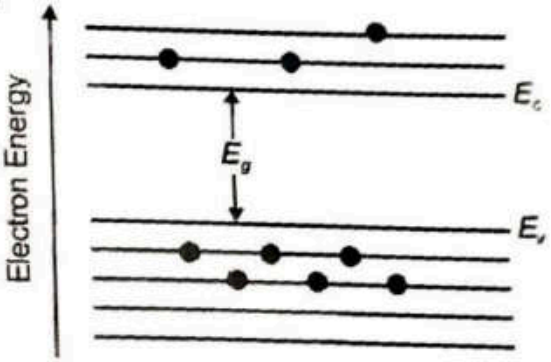
(4) 7 s

7. The correct energy band representation of intrinsic semiconductor at temperature 0 K is
(o → holes
• → electrons)

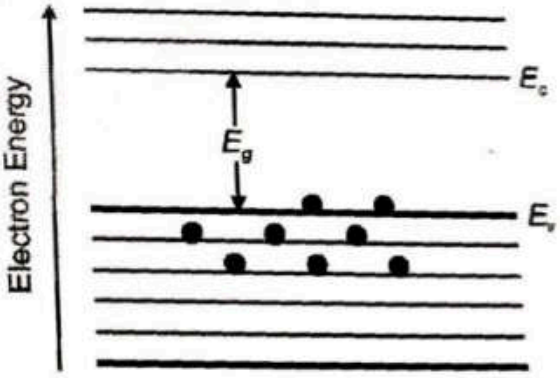
(1)



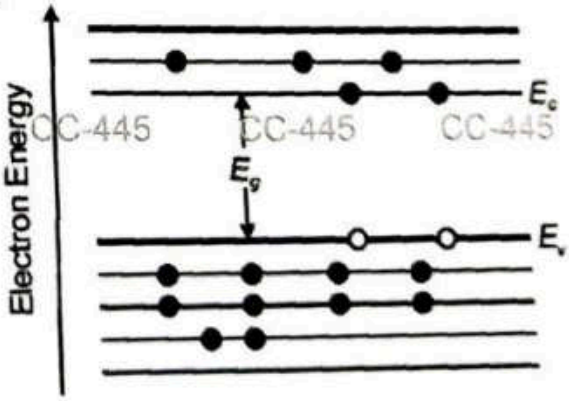
(2)



(3)

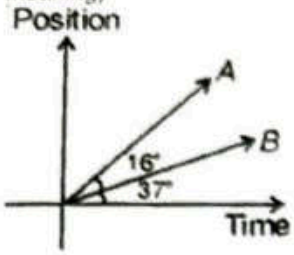


(4)



Vivek Krishna
CC-445

8. The position-time graph for two particles A and B are as follows. The ratio of $\frac{v_A}{v_B}$ is (All symbols have their usual meaning)



(1) 1

(2) $\frac{4}{3}$

(3) $\frac{16}{9}$

(4) $\frac{3}{4}$

9. Which of the following is/are not the property of nuclear force?

(1) Nuclear forces are non-central

(2) Nuclear forces do not obey inverse square law

(3) Nuclear forces are charge dependent

(4) All of these

10. If N_A is number density of acceptor atoms added and N_D is number density of donor atoms added to a semiconductor. If n and p are the number density of electrons and holes in it, then

(1) $N_A + p = N_D + n$

(2) $n + p = N_D + N_A$

(3) $N_A + n = N_D + p$

(4) $(N_A + N_D)^2 = p + n$

. Kinetic energy of an object moving along a straight line is given by $K = \frac{A\sqrt{x}}{x^2+B}$, where A and B are dimensional constants and x is position of object. The dimensional formula for $\frac{A}{B}$ is

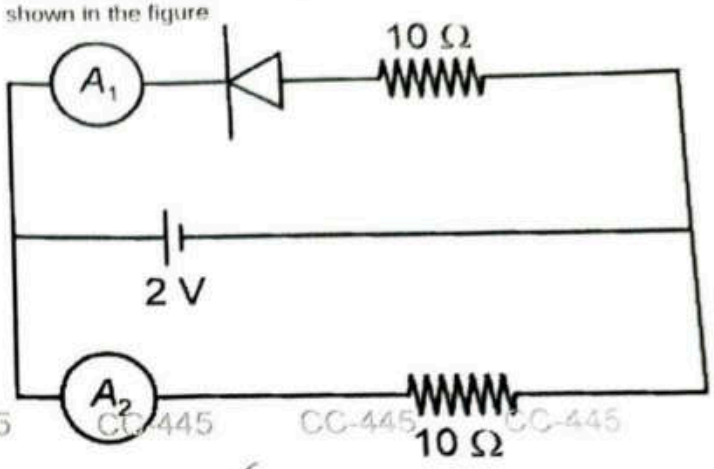
(1) $[M^1L^{7/2}T^{-2}]$

(2) $[M^{-1}L^{-3/2}T]$

(3) $[M^{-1}L^{-7/2}T^2]$

(4) $[ML^{3/2}T^{-2}]$

12. Find the respective readings of ideal ammeters A_1 and A_2 shown in the figure



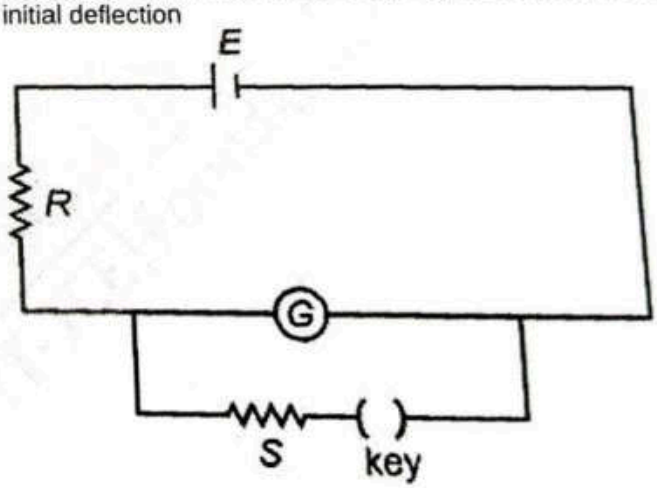
(1) Zero, 0.2 A

(2) 0.2 A, Zero

(3) Zero, Zero

(4) 0.2 A, 0.2 A

13. In the given circuit of half deflection, if E = 3 V, R = 2 kΩ and G = 100 Ω, then the current passing through shunt resistance when deflection of the galvanometer is half of initial deflection



(1) 0.75 mA

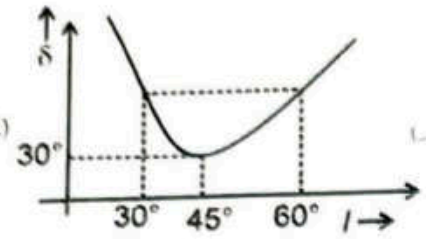
(2) 0.25 mA

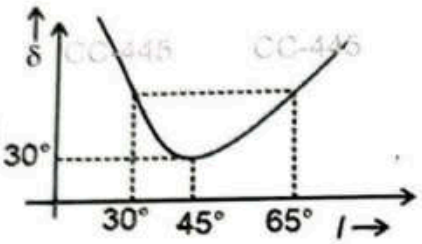
(3) 0.5 mA

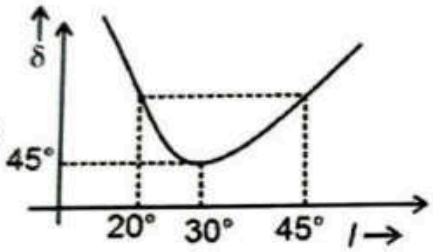
(4) 2.5 mA

CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445
Vivek Krishna
CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445

14. A laboratory supervisor observes experimental observations plotted by three different students performing the experiment for variation of angle of deviation of an equilateral prism with angle of incidence. The correct graph among these are (if prism is an equilateral one)

(1)


(2)


(3)


(4) All are correct

15. Consider the following statement.
Statement A : The dimensional correctness of an equation can be verified using the principle of homogeneity.
Statement B : All unitless quantities are dimensionless.
Which option is correct based an above information?

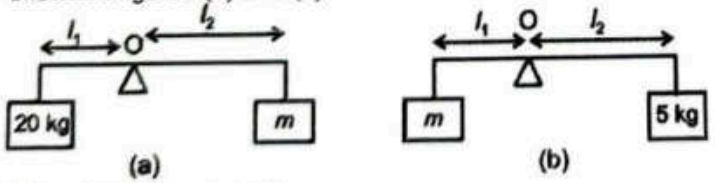
(1) Both statements A and B are true

(2) Both statements A and B are false

(3) Statement A is true but statement B is false

(4) Statement A is false but statement B is true

16. In an experiment with a beam balance an unknown mass m is balanced by two known masses of 20 kg and 5 kg as shown in figures (a) and (b).



The unknown mass m is

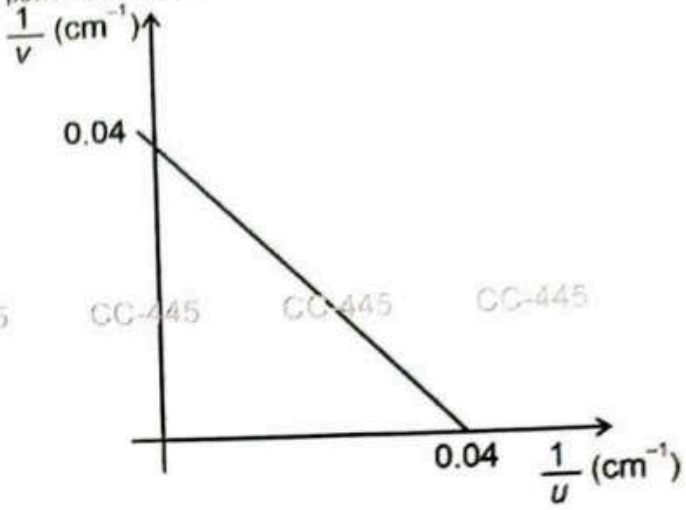
(1) 4 kg

(2) 8 kg

(3) 12 kg

(4) 10 kg

17. The focal length (f) of a thin convex lens is determined by two-pin method. The numerical values of the image-distance (v) are measured by various object distances (u). The given figure shows a graph of $\frac{1}{v}$(in cm^{-1}) versus $\frac{1}{u}$(in cm^{-1}). The power of the lens is



(1) 1 D

(2) 2 D

(3) 4 D

(4) 5 D

18. If m, m_n and m_p are the masses of $_ZX^A$ nucleus, neutron and proton respectively, then

(1) $m = (A - Z)m_n + Zm_p$

(2) $m > (A - Z)m_n + Zm_p$

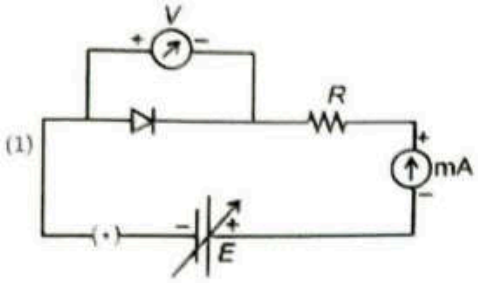
(3) $m < (A - Z)m_n + Zm_p$

(4) $m = (A - Z)m_p + Zm_n$

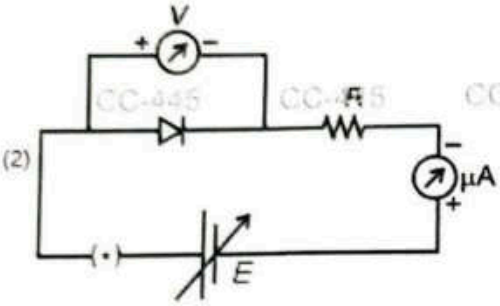
Vivek Krishna
CC-445

19. Which of the following diagram correctly study *I-V* characteristics curve for P-N junction in forward bias?

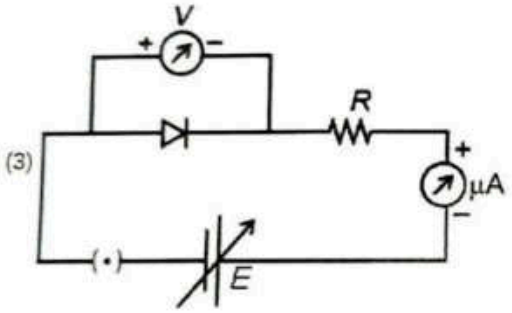
(1)



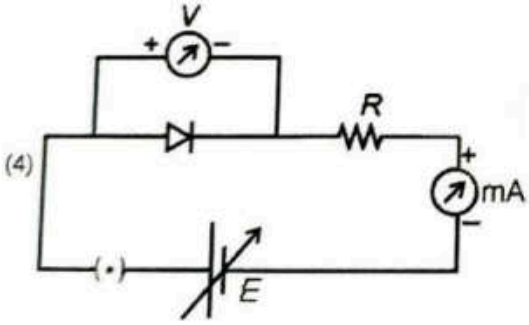
(2)



(3)



(4)



20. The experiment which is used to determine Young's modulus of the material of a given wire, is

(1) Resonance tube experiment

(2) Displacement method

(3) Searle's experiment

(4) Young's double slit experiment

21. A small steel ball of radius r is allowed to fall under gravity through a column of viscous liquid of coefficient of viscosity η. After some time the velocity of the ball attains a constant speed called terminal velocity. The coefficient of viscosity depends upon (i) (m) mass of the ball (ii) v_T (iii) g (iv) r (radius of ball).
Which of the following relations is dimensionally correct?

(1) $\eta \propto \frac{mg}{(v_T)r}$

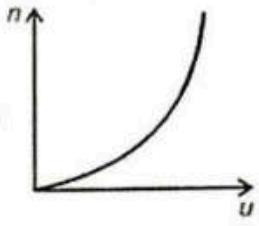
(2) $\eta \propto \frac{(v_T)r}{mg}$

(3) $\eta \propto \frac{rmg}{(v_T)}$

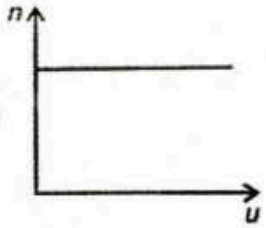
(4) $\eta \propto mgr$

22. A physical quantity is measured and its value is found to be nu, where n = numerical value and u = unit. Then which of the following graph is correct?

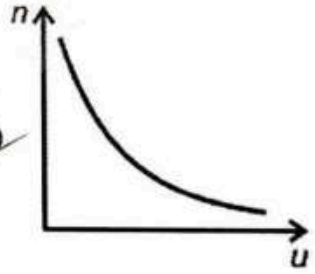
(1)



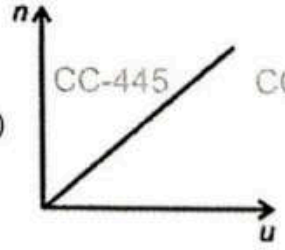
(2)



(3)



(4)



Vivek Krishna

23. In a mixture of H-Li^{+2} gas, H atoms and Li^{+2} ions are excited to their respective second excited states. Subsequently, each of the H atoms transfer their total excitation energy to each of the Li^{+2} ions (by collisions). Assuming the Bohr model of atom to be valid, the total energy of a Li^{+2} ion after reaching its' final quantum number, will be

(1) –0.54 eV

(2) –0.85 eV

(3) –1.51 eV

(4) –3.4 eV

24. The correct statement about radioactive decay among the following is

(1) In a γ-decay, the element does not change

(2) In an α-decay, the helium nucleus is emitted

(3) In a β^+ decay, positron is emitted from the nucleus

(4) All of these

25. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A): In a spontaneous fission reaction, when a heavy nucleus is split into two medium sized ones, then there is release of energy.
Reason (R): The original nucleus has more binding energy per nucleon than each of the two new product nuclei.
In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

26. A measurement of 4.645 m after rounding-off to three significant figures, will become

(1) 4.64 m

(2) 4.60 m

(3) 4.65 m

(4) 4.70 m

27. A particle of mass m moves in a circular orbit (under the influence of a central force) with potential energy $U(r) = \frac{cr^4}{4}$, where c is a positive constant and r is the radial distance from the origin. Assuming the Bohr's quantisation rule to be valid, the total energy of the particle will be

(1) $\frac{1}{3}\left(\frac{n^{10}h^{10}}{1024\pi^{10}m^5c^2}\right)^{1/3}$

(2) $\frac{1}{3}\left(\frac{n^6h^6c^2}{64\pi^6m^3}\right)^{1/6}$

(3) $\frac{5}{6}\left(\frac{n^6h^6c^2}{64\pi^6m^3}\right)^{1/6}$

(4) $\frac{5}{6}\left(\frac{n^{10}h^{10}}{1024\pi^{10}m^5c^2}\right)^{1/3}$

28. A ball is thrown vertically upward from the ground and it returns back to its initial position after 5 seconds. Which of the following quantity for the ball, has become zero at the end of 5 seconds?

(1) Path length

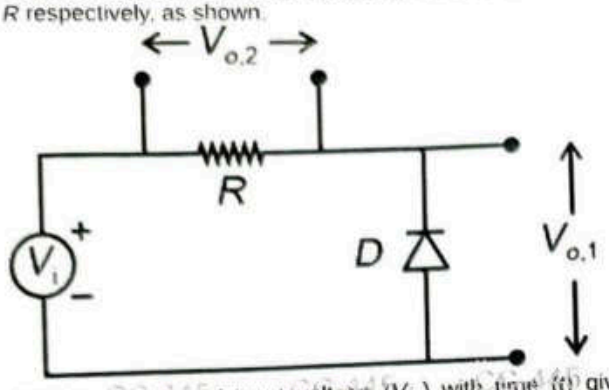
(2) Average speed

(3) Displacement

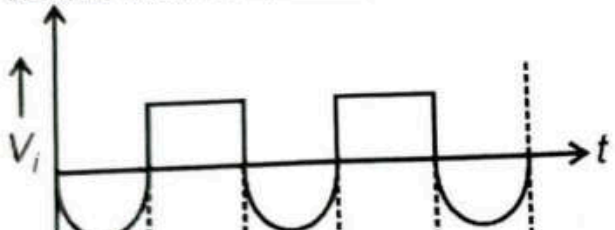
(4) Both (1) and (2)

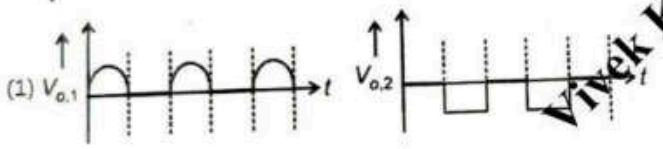
CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445
Vivek Krishna
CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445

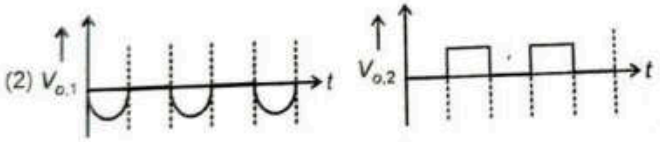
29. In a half-wave rectifier, output voltages $V_{o\ ,1}$ and $V_{o\ ,2}$ are obtained across the ends of ideal diode D and load resistor R respectively, as shown.

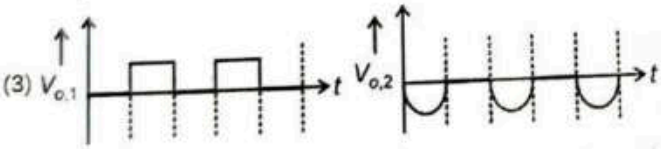


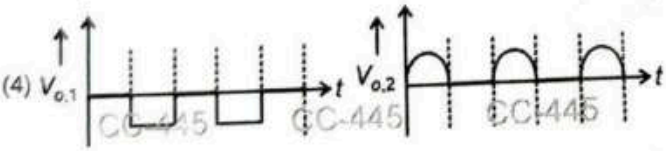
For the variation of input voltage (V_i) with time (t) given below, the variation of output voltages will be



(1) 

(2) 

(3) 

(4) 

30. If M_X and M_Y denote the atomic masses of parent and daughter nuclei respectively in a radioactive β^- decay, then Q-value of this nuclear reaction will be (where symbols have their usual meaning)

(1) $(M_X - M_Y)c^2$

(2) $(M_X - M_Y - m_e)c^2$

(3) $(M_X - M_Y - 2m_e)c^2$

(4) $(M_X - M_Y + 2m_e)c^2$

31. The resistance of a thin wire is given as $R = \frac{\rho l}{A}$, where l is length, A is cross sectional area and ρ is resistivity. If measured value of length $l = (10.0 \pm 0.1)$ cm, radius of wire $r = (1.0 \pm 0.02)$ cm and resistance $R = (20.0 \pm 0.5)\ \Omega$, then maximum percentage error in its resistivity is

(1) 6%

(2) 3%

(3) 7.5%

(4) 4.5%

32. Velocity of an electron in second stable orbit of hydrogen atom is (where c is the speed of light)

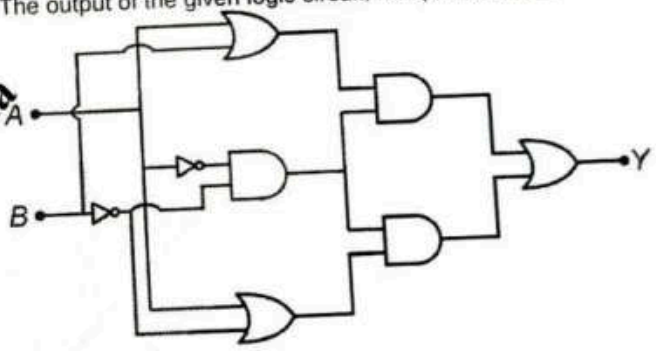
(1) $\frac{c}{224}$

(2) $\frac{c}{411}$

(3) $\frac{c}{274}$

(4) $\frac{c}{472}$

33. The output of the given logic circuit, is equivalent to a



(1) OR gate

(2) AND gate

(3) NOR gate

(4) NAND gate

34. In a p-n junction, if an electron from n-side approaches the p-n junction with speed 5.4×10^5 m/s, then it enters the p-side with speed 4.6×10^5 m/s.

If thickness of the depletion region is 5×10^{-8} m, then intensity of electric field in this region will be (take mass of electron as 9×10^{-31} kg, consider velocity drop only due to electric field and neglect any other loss at the junction)

(1) 2.25×10^7 V/m

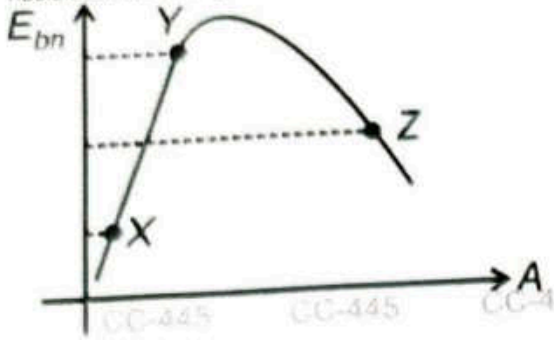
(2) 4.5×10^6 V/m

(3) 2.475×10^5 V/m

(4) 4.95×10^5 V/m

Vivek Krishna
CC-445

35. Binding energy per nucleon (E_{bn}) vs mass number (A) curve for nuclei is shown in the figure. Three nuclei X, Y, Z are indicated on the curve. The energy required to remove one neutron from its corresponding element of the three nuclei, can be arranged as (from lowest to highest)



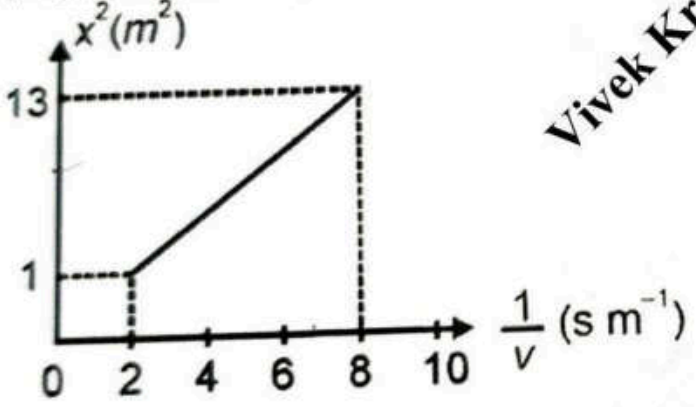
(1) $X < Y < Z$

(2) $Z < X < Y$

(3) $X < Z < Y$

(4) $Z < Y < X$

36. The graph of inverse of velocity $\left(\frac{1}{v}\right)$ vs square of position (x^2) for a particle moving in a straight line is as shown in the figure. The time taken by the particle to move from $x = 2$ m to $x = 3$ m, will be (in seconds)



(1) 3

(2) $\frac{11}{3}$

(3) $\frac{16}{3}$

(4) 4

37. Isotones are nuclides having the same number of

(1) Protons

(2) Neutrons

(3) Electrons

(4) Both (1) and (2)

38. The orbital angular momentum of an electron in a H-like atom is $\frac{2h}{\pi}$. The radius of its orbit is $3.2a_0$. The wavelength of emitted photon, when the atom de-excites to ground state is (where symbols have their usual meaning)

(1) $\frac{36}{5R}$

(2) $\frac{16}{9R}$

(3) $\frac{16}{375R}$

(4) $\frac{50}{36R}$

39. Ratio of nuclear radius of ^{27}Al and that of ^{125}Fe is

(1) 1 : 1

(2) 1 : 3

(3) 3 : 5

(4) 27 : 125

40. For a particle undergoing uniform motion along a straight line, the correct statement is:
(A) Its position-time graph is a straight line.
(B) Its velocity-time graph is a straight line.
(C) Its acceleration-time graph is a straight line.
Choose the correct answer among the options given below.

(1) (A) only

(2) (B) Only

(3) (B) and (C) only

(4) All (A), (B) and (C)

41. A particle moves along y-axis with acceleration $a = (4t - 12)$, where t is in seconds. The particle is initially at the origin and moves along positive y-axis with speed 16 m/s. Average speed of the particle in first five seconds will be

(1) $\frac{8}{3}$ m/s

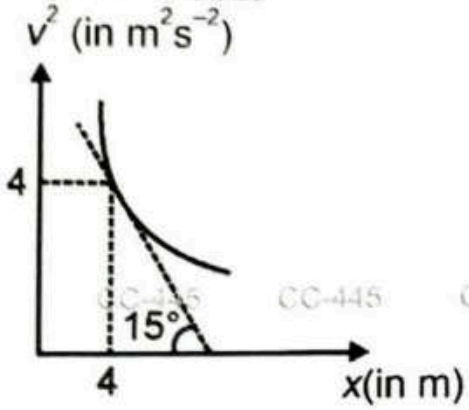
(2) $\frac{56}{15}$ m/s

(3) 3 m/s

(4) 4 m/s

Vivek Krishna
CC-445

42. A particle is moving along a straight line. The graph for its motion is as shown in the figure. The magnitude of acceleration of the particle when its position is at $x = 4$ m, will be (where v is velocity, x is position of particle and use $\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$)



(1) $\frac{1}{2}$ m s^{-2}

(2) $\left(1 - \frac{\sqrt{3}}{2}\right)$ m s^{-2}

(3) $(2 - \sqrt{3})$ m s^{-2}

(4) $(2 + \sqrt{3})$ m s^{-2}

43. A large platform is moving vertically upwards with a constant acceleration of 4 m/s^2. At time $t = 0$, a girl standing on the platform throws a ball vertically upwards with a relative speed of 4 m/s w.r.t. platform. At this instant, platform was moving upwards with a speed of 4 m/s. By the time the ball strikes the platform for the first time, the platform would have risen by

(1) $\frac{144}{49}$ m

(2) $\frac{352}{49}$ m

(3) $\frac{144}{7}$ m

(4) $\frac{352}{7}$ m

44. Which of the following nuclear models of hydrogen atom is/are stable?

(1) Thomson's model

(2) Rutherford's model

(3) Bohr's model

(4) All of these

45. Match column-I with column-II for hydrogen spectrum (where R is Rydberg's constant)

	Column-I (n^{th} line in)		Column-II (Corresponding wavelength)
a.	Balmer series	(i)	$\frac{16(n^2+8n+16)}{R(n^2+8n)}$
b.	Paschen series	(ii)	$\frac{(n^2+2n+1)}{R(n^2+2n)}$
c.	Brackett series	(iii)	$\frac{4(n^2+4n+4)}{R(n^2+4n)}$
d.	Pfund series	(iv)	$\frac{9(n^2+6n+9)}{R(n^2+6n)}$
		(v)	$\frac{25(n^2+10n+25)}{R(n^2+10n)}$

Choose the correct answer from the options given below.

(1) a(iii), b(ii), c(i), d(iv)

(2) a(ii), b(iii), c(iv), d(i)

(3) a(iii), b(iv), c(i), d(v)

(4) a(iv), b(ii), c(i), d(iii)

CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445

CHEMISTRY

46. What will be the work function of the metal (in eV), if energy of incident light is 6 eV and kinetic energy of photoelectron emitted is 3.2×10^{-19} J?

(1) 0.5 eV

(2) 4 eV

(3) 2 eV

(4) 1.5 eV

47. If 14 g of nitrogen gas is mixed with 6 g of hydrogen gas, then the maximum mass of ammonia produced in the reaction is

(1) 17 g

(2) 15 g

(3) 20 g

(4) 12 g

48. Benzylamine and aniline can be chemically distinguished by

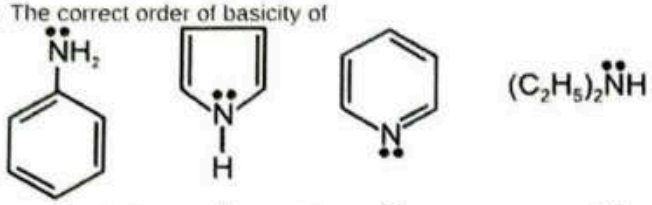
(1) Hinsberg's reagent

(2) Br_2/water

(3) Dilute H_2SO_4

(4) $CHCl_3|KOH|\Delta$

49. The correct order of basicity of



I II III IV

in water is

(1) IV > III > I > II

(2) IV > I > III > II

(3) IV > II > III > I

(4) III > I > IV > II

50. Mass of $CaCO_3$ required to react completely with 320 mL of 0.25 M HCl is

$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$

(1) 4 g

(2) 8 g

(3) 6.5 g

(4) 12.5 g

51. Which one of the following is least likely to exist?

(1) HOF

(2) $HClO_4$

(3) HIO_3

(4) $HBrO_2$

52. Match List I with List II.

	List I Vitamins		List II Deficiency disease
(a)	Riboflavin	(i)	Convulsions
(b)	Pyridoxine	(ii)	Beri-Beri
(c)	Ascorbic acid	(iii)	Cheilosis
(d)	Thiamine	(iv)	Scurvy

Choose the correct option.

(1) (a)–(i), (b)–(ii), (c)–(iii), (d)–(iv)

(2) (a)–(iv), (b)–(iii), (c)–(ii), (d)–(i)

(3) (a)–(iii), (b)–(iv), (c)–(i), (d)–(ii)

(4) (a)–(iii), (b)–(i), (c)–(iv), (d)–(ii)

53. Given below are two statements.

Statement-I: Mathematically, Heisenberg's uncertainty principle can be given as $\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$

Statement-II: The effect of Heisenberg uncertainty principle is significant only for motion of microscopic objects and is negligible for that of macroscopic objects.

In the light of above statements select the correct option.

(1) Statement I is incorrect but statement II is correct

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

54. IF_7 on complete hydrolysis gives

(1) HOF and HIO_3

(2) HF and HIO_3

(3) HOF and HIO_4

(4) HF and HIO_4

55. Maximum number of electrons in a subshell with $n = 2$ and $l = 1$ is

(1) 2

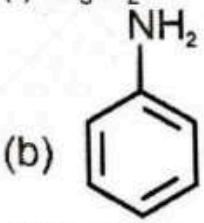
(2) 6

(3) 10

(4) 14

56. The amines which cannot be synthesised by using Gabriel phthalimide synthesis are

(a) CH_3NH_2



(b)

(c) $CH_3CH_2NH_2$

(d) $(CH_3)_3N$

Choose the correct option

(1) (a), (b) and (c) only

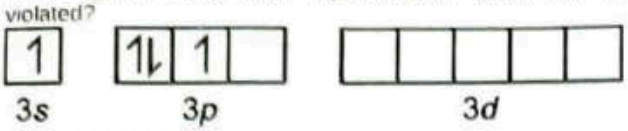
(2) (a) and (b) only

(3) (c) and (d) only

(4) (b) and (d) only

Vivek Krishna

57. In the given configuration representation which rule is violated?



(A) Aufbau principle
(B) Pauli's exclusion principle
(C) Hund's rule of maximum multiplicity
Choose the correct option.

(1) (A), (B) and (C)

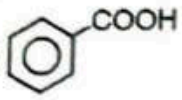
(2) (A) and (C) only

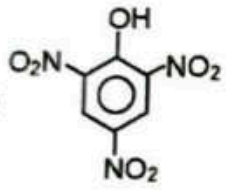
(3) (B) and (C) only

(4) (B) only

58. The compound which does not produce effervescence with sodium bicarbonate is

(1) CH_3COOH

(2) 

(3) 

(4) $CH_3CH_2 - OH$

59. The correct order of bond dissociation enthalpy of halogens is

(1) $F_2 > Cl_2 > Br_2 > I_2$

(2) $Cl_2 > Br_2 > F_2 > I_2$

(3) $F_2 > Br_2 > I_2 > Cl_2$

(4) $F_2 > Cl_2 > I_2 > Br_2$

60. Match List-I with List-II.

List-I Compounds		List-II Colour
a. $HgO{\cdot}Hg(NH_2)I$	(i)	White
b. $[Cu(NH_3)_4]SO_4$	(ii)	Red
c. $[Ce(NO_3)_4(ROH)_3]$	(iii)	Brown
d. $Mg(NH_4)PO_4$	(iv)	Deep blue

Choose the correct option.

(1) a(iii), b(iv), c(ii), d(i)

(2) a(iv), b(iii), c(i), d(ii)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(i), b(iv), c(ii), d(iii)

61. If 6.022×10^{22} molecules of glucose are dissolved in 500 g of water, then the molality of this solution is

(1) 0.2 m

(2) 0.1 m

(3) 1 m

(4) 0.5 m

62. Given below are the two statements
Statement I: SO_2 is a colourless gas and is highly soluble in water.
Statement II: SO_2 reacts with chlorine in the presence of charcoal to give thionyl chloride.
In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are correct.

(4) Both statement I and statement II are incorrect.

63. Lactose on hydrolysis gives

(1) Galactose and Fructose

(2) Galactose and Glucose

(3) Galactose and Ribose

(4) Glucose and Fructose

64. Colour of phenolphthalein in acidic medium is

(1) Pink

(2) Orange

(3) Blue

(4) Colourless

65. Given below are two statements.
Statement I : When nucleoside is linked to phosphoric acid at 3′-position of sugar moiety we get nucleotide.
Statement II : Nucleotides are joined together by phosphodiester linkage between 5′ and 3′ carbon atoms of the pentose sugar.
In the light of above statements choose the correct option.

(1) Statement I is incorrect but statement II is correct

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

Vivek Krishna
CC-445

66. Match List-I with List-II

	List-I (Oxides of Nitrogen formed in the reaction)		List-II (Physical appearance and chemical nature of the oxide of nitrogen formed)
a.	$NO + N_2O_4 \xrightarrow{250K}$	(i)	Colourless gas, neutral
b.	$Pb(NO_3)_2 \xrightarrow{673K}$	(ii)	Brown gas, acidic
c.	$HNO_3 + P_4O_{10} \rightarrow$	(iii)	Blue solid, acidic
d.	$NH_4NO_3 \xrightarrow{Heat}$	(iv)	Colourless solid, acidic

Choose the correct option

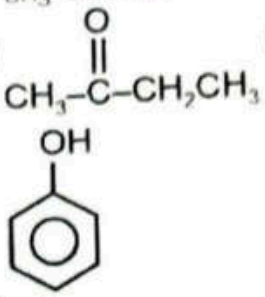
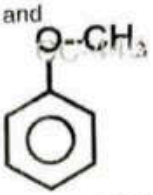
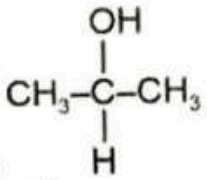
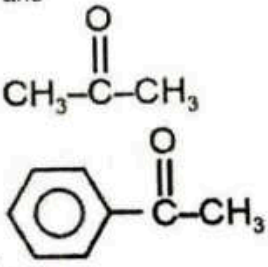
(1) a(i), b(ii), c(iii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(ii), c(iv), d(i)

(4) a(ii), b(iii), c(iv), d(i)

67. Match the compounds given in column I with reagents (which can distinguish the compounds) given in column II

Column I	Column II
CH_3–CHO and (a) 	(i) Neutral $FeCl_3$
(b) and 	(ii) Lucas reagent
(c)  and 	(iii) Tollens' reagent
(d) and	(iv) I_2/NaOH

Select the correct option.

(1) a(iii), b(ii), c(i), d(iv)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(ii), b(iii), c(iv), d(i)

(4) a(iv), b(i), c(iii), d(ii)

68. Suppose the elements X and Y combine to form two compounds X_2Y and X_3Y_4. 0.1 mole of X_2Y weighs 10 g and 0.05 mole of X_3Y_4 weighs 15 g, the atomic masses of X and Y respectively are

(1) 20 u and 60 u

(2) 50 u and 30 u

(3) 60 u and 20 u

(4) 30 u and 50 u

Vivek Krishna

CC-445

69. Number of moles of $KMnO_4$ required to react completely with 1 mole of $FeSO_4$ in acidic medium is

(1) 5

(2) $\frac{1}{5}$

(3) 3

(4) $\frac{1}{3}$

70. Volume of CO_2 obtained at STP on complete combustion of 87 g n-butane in presence of excess oxygen gas is

(1) 112 L

(2) 156.8 L

(3) 201.6 L

(4) 134.4 L

71. The ratio of the wavelengths for the longest wavelength transition in the Balmer series to the shortest wavelength transition in the Paschen series of atomic hydrogen is

(1) 4 : 5

(2) 7 : 20

(3) 7 : 36

(4) 9 : 20

72. Match list-I with list-II

	List-I Amount of substance		List-II Number of atoms
a.	6.6 g of CO_2	(i)	0.8 N_A
b.	2.7 g of $C_6H_{12}O_6$	(ii)	0.6 N_A
c.	2.24 L of C_2H_6 at STP	(iii)	0.45 N_A
d.	3.36 L of NH_3 at STP	(iv)	0.36 N_A

Choose the most appropriate answer from the options given below

(1) a(iii), b(iv), c(i), d(ii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iii), b(i), c(iv), d(ii)

73. If a particle of mass 0.2 mg is moving with a velocity of 4000 ms^{-1} then the de-Broglie wavelength associated with the moving particle will be ($h = 6.6 \times 10^{-34}$ Js)

(1) 8.25×10^{-31} m

(2) 8.25×10^{-34} m

(3) 8.25×10^{-32} m

(4) 8.25×10^{-33} m

74. Consider the following statements

(a) Zinc on reaction with dilute HNO_3 liberates N_2O gas

(b) White phosphorous disproportionates to phosphine and sodium hypophosphite when boiled with concentrated NaOH solution in an inert atmosphere of CO_2

(c) PCl_3 hydrolyses in presence of moisture giving fumes of HCl

(d) Orthophosphorous acid on heating disproportionates to hypophosphorous acid and orthophosphoric acid

The correct statements are

(1) (b) and (c) only

(2) (b), (c) and (d) only

(3) (a), (b), (c) and (d)

(4) (a), (b) and (c) only

75. Given below are the two statements

Statement-I: HCl is a stronger acid than HF.

Statement-II: Bond dissociation enthalpy of HCl is lesser than HF.

In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

76. Correct order of the boiling point of the given compounds is

(1) $BiH_3 > SbH_3 > NH_3 > AsH_3 > PH_3$

(2) $NH_3 > BiH_3 > SbH_3 > AsH_3 > PH_3$

(3) $NH_3 > BiH_3 > SbH_3 > PH_3 > AsH_3$

(4) $BiH_3 > NH_3 > SbH_3 > AsH_3 > PH_3$

77. Consider the following statements

(a) White phosphorus is a translucent white waxy solid.

(b) Red phosphorus glows in dark.

(c) White phosphorus consists of discrete tetrahedral P_4 molecule.

(d) Red phosphorus possesses iron grey lustre.

The correct statements are

(1) (a), (b) and (c) only

(2) (a), (c) and (d) only

(3) (b), (c) and (d) only

(4) (a) and (d) only

78. Given below are the two statements

Statement-I: Sucrose is dextrorotatory but after hydrolysis gives dextrorotatory glucose and laevorotatory fructose.

Statement-II: Sucrose reduces Tollens' reagent.

In light of above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is incorrect but statement II is correct

(4) Statement I is correct but statement II is incorrect

Vivek Krishna

79. The pair which represents non-essential amino acids is

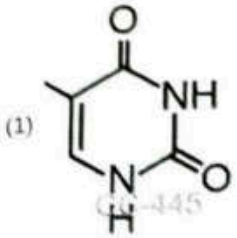
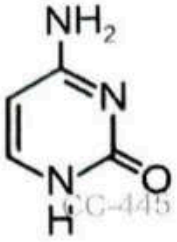
(1) Proline and serine

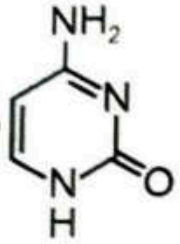
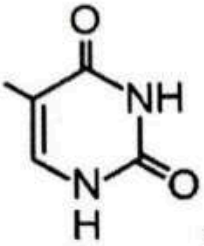
(2) Lysine and Histidine

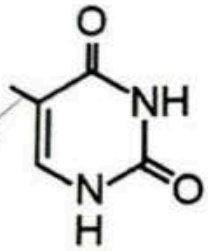
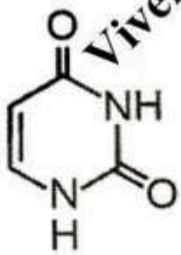
(3) Leucine and Valine

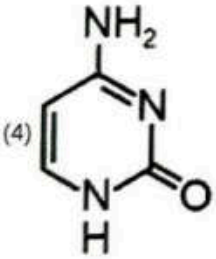
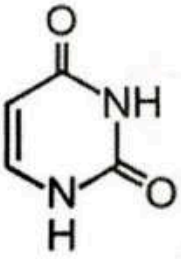
(4) Methionine and phenylalanine

80. Structures of thymine and uracil respectively are

(1)  and 

(2)  and 

(3)  and 

(4)  and 

Vivek Krishna

81. Consider the following statements

(a) The pentaacetate of glucose does not react with hydroxylamine.

(b) Glucose on oxidation with nitric acid yields saccharic acid.

(c) α-D-(–)-Fructofuranose and β-D-(–)-Fructofuranose are non-superimposable mirror images of each other.

(d) Maltose is composed of two α-D-Glucose units.

The correct statements are

(1) (a), (b) and (c) only

(2) (a), (b) and (d) only

(3) (a), (b), (c) and (d)

(4) (b) and (d) only

82. Consider the following compounds

(a) N-Methylmethanamine

(b) N, N-Dimethylmethanamine

(c) Phenylmethanamine

(d) Methanamine

The correct order of pK_b values of the given amines in aqueous medium is

(1) (c) > (b) > (d) > (a)

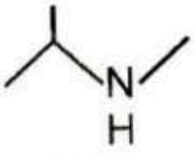
(2) (b) > (c) > (d) > (a)

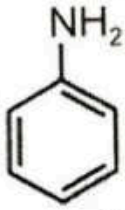
(3) (a) > (d) > (b) > (c)

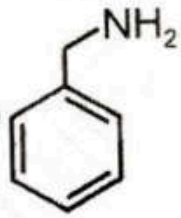
(4) (b) > (c) > (a) > (d)

83. Consider the following compounds

(i) 

(ii) 

(iii) 

(iv) 

The compound(s) which give positive isocyanide test is/are

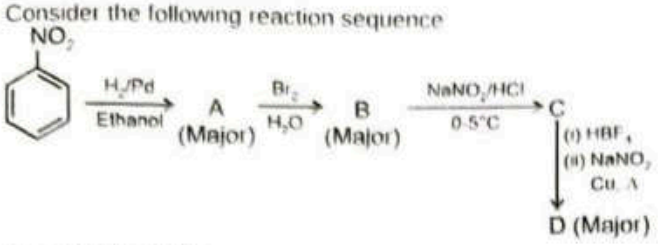
(1) (ii) and (iii) only

(2) (i) only

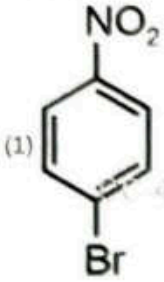
(3) (i), (iii) and (iv) only

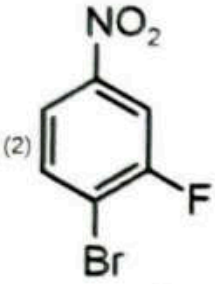
(4) (i) and (ii) only

84. Consider the following reaction sequence

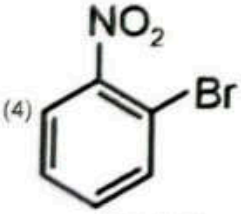


Major product D is

(1) 

(2) 

(3) 

(4) 

85. The compound which does not react with Hinsberg's reagent is

(1) Propan-2-amine

(2) Ethanamine

(3) N-Methylethanamine

(4) N, N-Dimethylmethanamine

86. If mass percentages of carbon, hydrogen and oxygen in an organic compound are 54.54%, 9.1% and 36.36% respectively then the empirical formula of the compound will be

(1) CHO_2

(2) C_2H_4O

(3) CH_4O

(4) CHO

87. Match list-I with list-II

	List-I Compound		List-II Preparation method
a.	Metaphosphoric acid	(i)	$P_2O_3 + H_2O$
b.	Phosphonic acid	(ii)	Red P_4 + alkali
c.	Phosphinic acid	(iii)	White P_4 + alkali
d.	Hypophosphoric acid	(iv)	Phosphorus acid + Br_2, heat in a sealed tube

Choose the most appropriate answer from the options given below

(1) a(iv), b(i), c(ii), d(iii)

(2) a(iv), b(i), c(iii), d(ii)

(3) a(i), b(iv), c(iii), d(ii)

(4) a(ii), b(i), c(iv), d(iii)

88. Consider the following statements

(a) In Deacon's process, the catalyst used is $CuCl_2$

(b) Chlorine on reaction with hot and concentrated NaOH gives NaOCl as one of the products

(c) Iodine is oxidised to iodic acid with chlorine in aqueous medium

(d) Chlorine can be used in sterilising drinking water.

The correct statements are

(1) (b) and (d) only

(2) (a), (b) and (c) only

(3) (a), (b), (c) and (d)

(4) (a), (c) and (d) only

Vivek Krishna

CC-445

C 54.54 4.54 2
H 9.1 9.1 4
O 36.36 2.02 1

4.54
2.02

89. Given below are the two statements

Statement-I: XeF_6 on reaction with RbF gives $Rb[XeF_7]$

Statement-II: XeF_4 on complete hydrolysis liberates F_2 gas.

In light of above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Both statement I and statement II are correct

(3) Both statement I and statement II are incorrect

(4) Statement I is incorrect but statement II is correct

90. Group reagents for Mn^{2+} and Sr^{2+} in qualitative analysis of cations respectively are

(1) H_2S in presence of dil. HCl and NH_4OH in presence of NH_4Cl

(2) Dilute HCl and $(NH_4)_2CO_3$ in presence of NH_4OH

(3) H_2S in presence of NH_4OH and dilute HCl

(4) H_2S in presence of NH_4OH and $(NH_4)_2CO_3$ in presence of NH_4OH

Vivek Krishna

BOTANY

91. The group of eukaryotic organisms having highest biodiversity in Amazonian rain forest, includes the organisms which

a. Help in breakdown of complex organic matter through mineralisation.

b. Belong to the order Diptera and show higher species count than the organisms belonging to the order Carnivora.

c. Show cytokinesis in their cells by forming the precursor of the structure that glues different neighbouring cells together.

d. Possess a network of tiny tubular structures in their cell for the production of steroidal hormones.

e. Have double membrane bound organelles in the mesophyll.

How many of the above given statements are **correct**?

(1) Four

(2) Three

(3) Two

(4) One

92. Interphase can result in

(1) Restoring the nucleo-cytoplasmic ratio.

(2) Introduction of variations in the genetic material due to the activity of recombinase.

(3) Disturbance in the ratio between the nucleus and the cytoplasm.

(4) The immediate production of diploid daughter cells with identical genetic complement.

93. Identify the following statements as True (T) or False (F) and choose the **correct** option.

A. The least number of meiotic divisions required to form 100 pollen grains in a typical angiospermic plant is 125.

B. The time required by a human cell in culture, to prepare itself for cell division is more than 22 hours.

C. In all types of eukaryotic cells, vacuoles can occupy 90% of total volume of the cell.

D. 27 microtubules are present in the core of the structure that gives rise to basal body of locomotory structure in a eukaryotic cell.

	A	B	C	D
(1)	T	T	F	F
(2)	F	F	T	T
(3)	F	T	F	T
(4)	T	F	T	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

94. Which of the following organisms will not be considered while forming an ecological pyramid of biomass of pond ecosystem?

(1) Phytoplankton

(2) Zooplankton

(3) Algae

(4) Fungi

95. Observe the following events:
a. Formation of the site of active ribosomal RNA synthesis.
b. Formation of the cell plate in the centre to form two daughter cells.
c. Complete disappearance of perinuclear space.
d. Centrosome moving towards the pole.
e. Attachment of spindle fibres to the sites found on the primary constriction.
f. Splitting of primary constriction.
Select the option depicting the **correct** sequence of events that occur during the formation of liquid endosperm in coconut.

(1) d → c → e → f → a → b

(2) c → e → f → d → b

(3) e → f → d → c → a

(4) c → e → f → a

96. Meiosis I is initiated after the

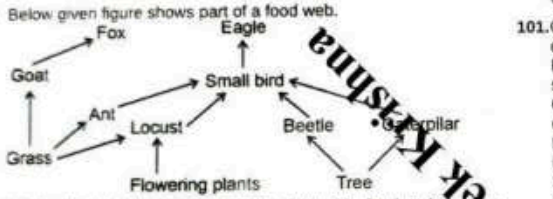
(1) Synaptonemal complex gets dissolved

(2) Separation of homologous chromosomes

(3) Parental chromosomes have replicated

(4) The completion of intrameiotic interphase

97. Below given figure shows part of a food web.



Which of the following statements about this food web is **correct**?

(1) If 20 J of energy is trapped at producer level, then the difference between the energy, available to fox and eagle would be 0.02 J.

(2) If the net productivity at the level of locust is 10x (K cal m^{-2}) yr^{-1} then the gross secondary productivity of the third trophic level in the food chain having flowering plants as producers would be x/10 (K cal m^{-2}) yr^{-1}.

(3) The standing crop at the IInd tropic level is more than the standing crop at the I^{st} trophic level.

(4) The food chain involving beetle will show spindle shaped pyramid of number

98. In bacteria, the small circular DNA outside the genomic DNA

(1) is immediately enclosed within a membrane

(2) Confers unique phenotypic characters

(3) Directs the genomic DNA replication

(4) Increases the surface area for metabolic activities

99. Certain discoveries are mentioned below:
a. Omnis *cellula-e cellula*
b. Observation of densely stained reticular structures near the nucleus
c. Proposition of fluid mosaic model
d. Discovery of the electron microscope
e. Proposition that animal cells had a thin outer layer known as plasma membrane
Select the option that chronologically define the timeline of these discoveries.

(1) a → e → d → b → c

(2) a → b → e → c → d

(3) b → d → a → e → c

(4) e → a → b → d → c

100. Which of the following cellular components is common between mycoplasma and a mesophyll cell?

(1) The structure which serves as the main arena of cellular activities

(2) Protein synthesising machinery with 60 S and 40 S subunits

(3) Photosynthetic pigment bearing organelle

(4) Sausage shaped organelle

101. Given below are some statements about the components of endomembrane system.
P. Golgi apparatus is extensive in cells specialised for secretion.
Q. Products of ER are modified during their transit from convex to concave face of Golgi apparatus.
R. The reticular structures of endomembrane system remain in close association.
S. The Golgi apparatus principally performs the function of modifying and packaging materials in entirely different but interconnected cis and convex faces, to be delivered to inter and intra-cellular targets.

(1) P and S are true and P is the reason for S

(2) P and R are true and P is the reason for R

(3) Q and R are true and Q is the reason for R

(4) Q and S are true and S is the reason for Q

102. How many of the following pairs is/are **correctly** matched?
a. Recombination nodule – Site at which crossing over occurs between sister chromatids
b. Completion of prophase – Marked by formation of mitotic apparatus
c. Zygotene – Sister chromatids in each chromosome are identical
d. Kinetochore – Junction between DNA and tubulin protein
e. Bivalent – Can be clearly seen during metaphase

(1) Two

(2) Three

(3) Four

(4) One

Vivek Krishna

103. Given below are some statements about decomposition.
P. Colloidal amorphous substance formed during the process can be degraded by means of humification.
Q. Build up of organic material can be easily observed in the region located at 71°N latitude of earth.
R. Low temperature inhibits decomposition.
S. Bacterial and fungal enzymes degrade detritus into simpler inorganic substances.

(1) P and S are true and S is the reason of P

(2) P and Q are true and Q is the reason of P

(3) R and S are true and S is the reason of R

(4) Q and R are true and R is the reason of Q

104. Natural interconnections of food chains

(1) Is not accommodated in ecological pyramids

(2) Always begins with dead organic matter

(3) Is not found in natural ecosystems

(4) Does not accommodate herbivores

105. A cell divides every minute. It takes an hour to fill a beaker of 50 ml.
(a) How much time it would take to fill 25 ml beaker?
(b) If this cell continues to divide further, how much time it will take to fill the beaker of volume 200 ml?
Select the option that **correctly** answers (a) and (b).

	(a)	(b)
(1)	30 min	40 min
(2)	25 min	28 min
(3)	50 min	60 min
(4)	59 min	62 min

(1) (1)

(2) (2)

(3) (3)

(4) (4)

106. Read the following statements.
a. It needs a constant supply of energy to counteract the universal tendency towards increasing disorderliness.
b. Its source of energy is sun.
c. It has a characteristic physical nature which can be characterised by interaction between biotic and abiotic components.
d. It provide services such as purification of air.
These characteristics are **true** for

(1) Raw material for decomposition

(2) Functional unit of nature

(3) Reservoir of nutrients obtained during decomposition

(4) Heterogeneity found at the lowest level of biological organisation

107. Which of the following expressions is **correct**?

(1) India < Norway (in terms of ecosystem diversity)

(2) Crustaceans > Molluscs (in terms of proportionate number of species)

(3) Amphibians < Birds (in terms of their species number in Amazonian rain forest)

(4) National parks > Wildlife sanctuaries (in terms of number in India)

108. Match column-I with column-II and select the **correct** option.

	Column-I		Column-II
a.	Protein synthesising structure of cells having naked genetic material	(i)	Store reserve material in inclusion bodies
b.	Cells that can show three layered cell envelope	(ii)	Are the smallest cells
c.	Cellular organisms having nucleoid but lack cell wall	(iii)	Are associated with plasma membrane of the cell
d.	Cell having approximately 52% protein and 40% lipids in its plasma membrane	(iv)	7.0 µm in diameter

(1) a(iii), b(iv), c(i), d(ii)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(ii), d(iv)

CC-445 CC-445 CC-445

109. Observe the figures given below



A B

Figure A differs from figure B as

(1) The latter represents the organism which is not culturable under laboratory conditions

(2) The former represents the core of an organelle which is surrounded by cell membrane and are outgrowths of the cell wall

(3) The latter is found in large number in liquid endosperm of coconut

(4) The former possesses 20 microtubules running perpendicularly to the long axis.

110. Sub-metacentric chromosome differs from metacentric chromosome as

(1) The former possesses primary constriction

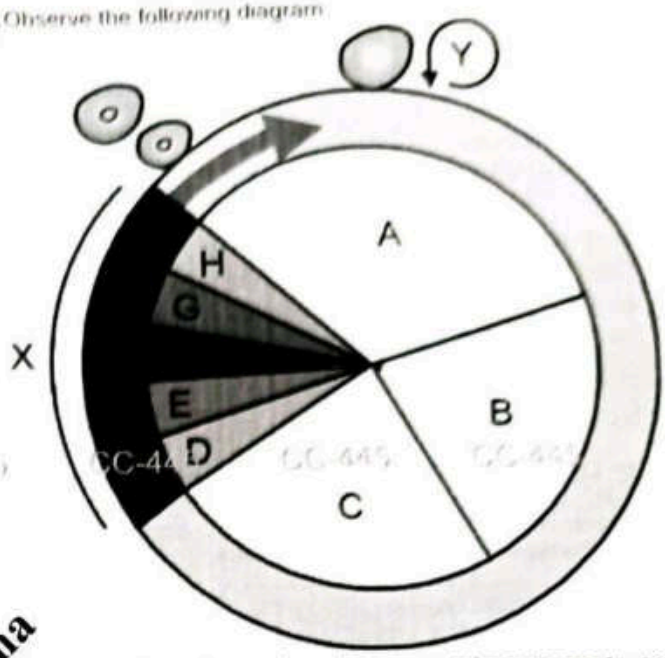
(2) In the latter the centromere is slightly away from the middle of the chromosome

(3) The former has distinguishable short and long arm

(4) The latter is not found in any eukaryotic cell

Vivek Krishna

111. Observe the following diagram



Based on the above given figure, certain statements are mentioned below:

(a) H is also a phase of karyokinesis that occurs during M-phase.

(b) Synthesis of polymers of nucleotides can occur only during the phase labelled as B.

(c) Transition from phase labelled as A to the one labelled as B is under genetic control.

(d) The time required to complete the phase labelled as X is less than 5 minutes in the yeast cell.

(e) In order to enter to the phase labelled as Y, the cell has to suspend its metabolic activity.

How many of them are **incorrect**?

(1) Two

(2) Three

(3) Four

(4) Five

112. Which of the following statements is **correct**?

(1) Herbivores are secondary consumers.

(2) The storage and movement of nutrient elements through the various components of the ecosystem is called energy flow.

(3) Secondary productivity is the rate of assimilation of food energy by the consumers.

(4) Organisms at each trophic level depend on those at the higher trophic level for their energy demands.

113. Read the following statements of Assertion (A) and Reason (R)

Assertion (A): The energy trapped by the producer is either passed on to a consumer or the organism dies.

Reason (R): All the energy that is trapped into an organism remains in it forever.

In the light of the above given statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

114. As per the careful analysis of the data proposed by credible agencies, it was proposed that 15,500 species world-wide are facing the threat of extinction. Below given are the groups of organisms that are facing this threat.

(a) Birds
(b) Mammals
(c) Amphibians
(d) Gymnosperms

Select the option that **correctly** describes the order in terms of percentage of threat of extinction of the species found in the groups given above.

(1) c = d > a > b

(2) d > c = b > a

(3) a > b > d > c

(4) c > d > b > a

115. How many of the following statements is/are **correct** for Western Ghats?

(a) This region is found in Karnataka and Madhya Pradesh.
(b) This region has lesser amphibian diversity than Eastern Ghats.
(c) It is a region of accelerated habitat loss and high endemism.
(d) Its ecosystem eventually becomes less functional if one of its tree frog species is lost forever.

(1) Four

(2) Three

(3) Two

(4) One

116. Read the following statements

a. The relation between species richness and area for bats is going to be a straight line on log scale.
b. Birds in California show steeper slope than frugivorous birds found in the tropical forests of Asia, if species-area relationship is plotted.
c. Sri Lanka is considered as a region of accelerated habitat loss.
d. India has more genetically different strains of rice than mango.

Select the **correct** ones:

(1) a, b and c only

(2) b, d and c only

(3) a, b and d only

(4) a, c and d only

117. How many of the following pairs are **incorrectly** matched?

a. Molluscs in New York – Show similar regression coefficient as shown by plants in Britain if species-area relationship is analysed.
b. Equador – Has lesser number of species of vascular plants than a forest of equal area in Midwest of the USA.
c. Brazil – The country in which 190 countries pledged their commitment in 1992 to achieve a significant reduction in the current rate of biodiversity.
d. India – Greater than 650 species are currently facing the threat of extinction, out of more than 15500 species world-wide.
e. Africa – Extinction of catfish (*Clarias gariepinus*) and cichlid fish.

(1) One

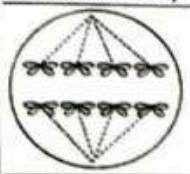
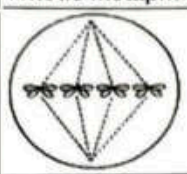
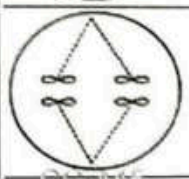
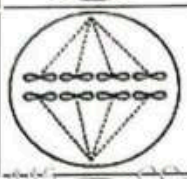
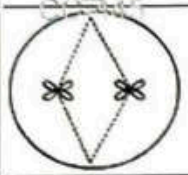
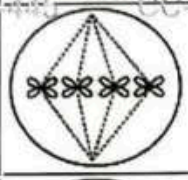
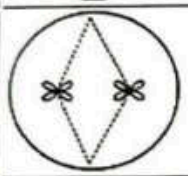
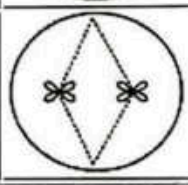
(2) Two

(3) Three

(4) Four

Vivek Krishna
CC-445

118. If an organism having 2n = 4 chromosomes, then which of the following options appropriately describes the meiotic II and mitotic metaphase?

	Meiotic II metaphase	Mitotic metaphase
(1)		
(2)		
(3)		
(4)		

(1) (1)

(2) (2)

(3) (3)

(4) (4)

119. Which of the following phases involves a major organisation of virtually all components of the cell?

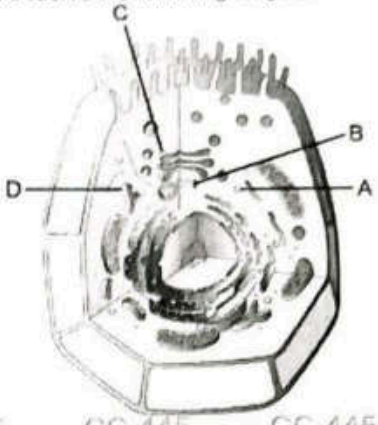
(1) G_1 phase

(2) S phase

(3) G_2 phase

(4) M phase

120. Observe the following diagram



Based on the above given diagram, match the labels given in column-I to their respective descriptions given in column II.

	Column-I		Column-II
(i)	A	P.	The central part of this structure is proteinaceous
(ii)	B	Q.	Major site for the synthesis of component which provides quasi-fluid nature to the plasma membrane
(iii)	C	R.	Membrane-bound minute vesicles that contain various enzymes, participate in photorespiration as well as present in both plant and animal cell
(iv)	D	S.	Densely stained reticular structures consisting of disc-shaped sacs

(1) (i)-R, (ii)-S, (iii)-P, (iv)-Q

(2) (i)-Q, (ii)-P, (iii)-R, (iv)-S

(3) (i)-R, (ii)-P, (iii)-S, (iv)-Q

(4) (i)-P, (ii)-Q, (iii)-S, (iv)-R

121. Cutting and clearing the Amazon rain forest for cultivating soyabean exemplifies which of the following 'evil quartet'?

(1) Habitat loss and fragmentation

(2) Co-extinction

(3) Alien species invasion

(4) Over-exploitation

122. Primary production is expressed in terms of

(1) $(kcal\ m^{-2})yr^{-1}$

(2) $kcal\ m^{-2}$

(3) $gm^{-2}\ yr^{-1}$

(4) $gm^{-2}\ yr^{-2}$

Vivek Krishna
CC-445

123. Read the following statements of **Assertion (A)** and **Reason (R)** and choose the **correct** option.
Assertion (A): Cell division stops in animals when the organisms become mature.
Reason (R): Unlike plants, animals do not show growth throughout their life cycle.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

124. The genetic variation shown by the medicinal plant *Rauwolfia vomitoria*, growing in different Himalayan ranges might be in terms of the potency and concentration of the

(1) Quinine

(2) Resins

(3) Dyes

(4) Reserpine

125. As per the IUCN 2004 red list, how many species of vertebrates were extinct in the last 500 years?

(1) 2000

(2) 338

(3) 359

(4) 784

126. __(A)__ first saw and described a live cell.
Select the **correct** option to fill (A).

(1) Antonie Von Leeuwenhoek

(2) Robert Brown

(3) Matthias Schleiden

(4) Theodore Schwann

127. In some bacteria, which of the following structures is known to help attach the bacteria to rocks in streams and also to the host tissues?

(1) Pili

(2) Flagella

(3) Fimbriae

(4) Cilia

128. Untangling of chromosomal material to make it condensed can be seen during

(1) Telophase

(2) Anaphase

(3) Metaphase

(4) Prophase

129. Given below are two food chains, in both of them 100 kJ of energy in the form of sunlight is falling on the green parts of the plant.
I. Plant → Grasshopper → Frog → Eagle
II. Plant → Grasshopper → Frog → Snake → Eagle
Based on the above information, select the **correct** statements.
a. These food chains exemplify the type of food chain, which forms the major conduit of energy flow in terrestrial ecosystem.
b. Secondary production of frog in food chain I is higher than the frog in food chain II.
c. The difference in amount of energy obtained by eagle in food chain I and in food chain II is 9×10^{-4} kJ.
d. All three types of pyramid for both food chains will be upright.

(1) a and c only

(2) a and b only

(3) b and d only

(4) c and d only

130. In which of the following phases nuclear membrane cannot be observed under microscope?

(1) Prophase

(2) Metaphase

(3) Late telophase

(4) S phase

131. In decomposition, complex organic compounds of detritus are converted to all of the following, **except**

(1) Starch

(2) Carbon dioxide

(3) Water

(4) Inorganic nutrients

132. Match column-I with column-II and select the **correct** option.

	Column-I		Column-II
a.	Stratification	(i)	Region of chromosome that can be easily stained by basic dyes
b.	Speciation	(ii)	Process by which homologous chromosomes are paired
c.	Satellite	(iii)	Process which is a function of time
d.	Synapsis	(iv)	Structural component of ecosystem

(1) a(iv), b(iii), c(ii), d(i)

(2) a(iii), b(iv), c(i), d(ii)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(iv), b(i), c(iii), d(ii)

CC-445 CC-445 CC-445

133. The **correct** expression of rate of production of organic matter by producers during photosynthesis per unit-time and area is

(1) GPP = NPP + R

(2) GPP = NPP – R

(3) GPP/NPP = R

(4) GPP = NPP × R

134. Which of the following regions is considered as the sacred groves in Meghalaya?

(1) Khasi and Jaintia Hills

(2) Aravalli Hills

(3) Chanda and Bastar

(4) Sarguja

CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445

135. Exploring molecular, genetic and species-level diversity for products of economic importance is termed as

(1) Bioremediation

(2) Bioprospecting

(3) Bioinformatics

(4) Biotechnology

Vivek Krishna

ZOOLOGY

136. rDNA technology refers to joining two different DNA molecules with an aim to 'X' the genes for various applications.
Here 'X' can be
(a) Isolate (b) Propagate
(c) Characterise (d) Manipulate
Select the most appropriate option.

(1) (a) only

(2) (c) and (d) only

(3) (d) only

(4) (a), (b), (c) and (d)

137. Consider the statements given below:
Statement A: Among prokaryotic hosts, the most widely used host is yeast, which was also used to produce the first human vaccine *via* RDT for hepatitis-B.
Statement B: Whenever a restriction enzyme cleaves at a site, it generates a 5'-phosphate group at one end of the break and a 3'-hydroxyl group at the other end of the break.
Select the most appropriate option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445

138. Plasmids act as vectors and they are not/do not

(1) Extra-chromosomal dsDNAs

(2) Occur in archaea and eukaryotes

(3) Ever contain genes that provide benefit to the host cell

(4) Always have a low copy number

139. Restriction endonucleases differ from exonucleases as

(1) Latter belongs to class III of enzymes

(2) Former cleaves the phosphodiester bond that directly joins the sugar residues of adjacent nucleosides in a DNA molecule

(3) Latter removes mononucleotides either from the 3' or 5' ends of a polynucleotide chain

(4) Former cuts at random sites within the DNA molecule

140. If a restriction enzyme 'X' cleaves a palindromic sequence in the manner as shown below, the type of ends generated will be

↓
5'-G A T A T C-3'
3'-C T A T A G-5'
↑

(1) Sticky ends

(2) Cohesive ends

(3) Staggered ends

(4) Blunt ends

141. How many of the enzymes mentioned in the box below is/are used in RDT?

DNA ligases, DNA polymerases, Reverse transcriptase, Nucleases

Choose the **correct** option.

(1) One

(2) Three

(3) Four

(4) Two

142. Select the odd one w.r.t. the physical methods used for introducing foreign DNA molecules into host cells.

(1) Micro-injection

(2) Gene gun

(3) Biolistics

(4) $CaCl_2$ treatment

143. A researcher is using pBR322 (4.36 kb) as a cloning vector to insert a 1.2 kb gene fragment at the *Pst* I site located within the plasmid. After ligation and transformation into *E.coli*, colonies are first selected on tetracycline and then replica-plated onto ampicillin.
To confirm successful cloning, plasmid DNA from several candidate colonies are digested with *Pst* I and analysed on a agarose gel.
Given the scenario, select the single phenotype-digest pattern that will be obtained for the non-recombinant transformants in the above case.

(1) tet^S, amp^R; single band will be seen on the gel

(2) tet^R, amp^R; two bands will be obtained on the gel

(3) tet^R, amp^R; single band will be obtained on the gel

(4) tet^S, amp^R; two bands will be obtained on the gel

144. The incorrect fact w.r.t. PCR is that it

(1) Involves the process of heating and cooling called thermal cycling

(2) Involves use of a thermostable DNA polymerase

(3) Involves the usage of two types of single-stranded synthetic oligonucleotide primers

(4) Is not a sensitive test and requires only a macro gram of the sample

145. **Assertion (A):** During the heat-shock step of *E.coli* transformation, cells are briefly exposed to elevated temperature (typically 92°C) to make the cell membrane temporarily permeable, allowing the plasmid DNA to enter the cell.
Reason (R): The transient heat shock causes partial denaturation of the bacterial cell wall, directly creating pores large enough for plasmid DNA to diffuse through by simple thermal motion.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true, (R) correctly explains (A)

(2) Both (A) and (R) are true, (R) does not explain (A) correctly

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

146. Match column I with column II w.r.t. important events in the field of biotechnology.

	Column I		Column II
a.	1972	(i)	First clinical gene therapy for ADA deficiency was performed
b.	1963	(ii)	Genetically engineered human insulin was made
c.	1983	(iii)	First artificial recombinant DNA molecule made
d.	1990	(iv)	The enzymes responsible for restricting the growth of bacteriophage in *E.coli* were isolated

Select the **correct** option.

(1) a(iii), b(ii), c(iv), d(i)

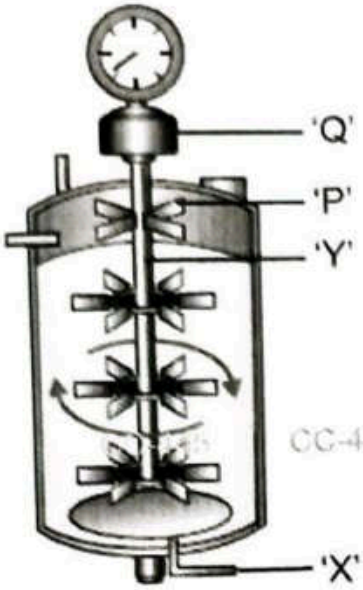
(2) a(i), b(iii), c(iv), d(ii)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(i), b(iii), c(ii), d(iv)

CC-445 CC-445 CC-445

147. Consider the image given below:



Identify the labelled parts and choose the **correct** option w.r.t. them.

(1) 'Y' represents a sparger that provides a regulated and sterile flow of air to the fermentation tank.

(2) 'P' is typically mounted on top of the fermenter to maximize foaming.

(3) 'Q' drives the vertical shaft bearing flat-bladded impellers to ensure bulk mixing and elevate oxygen transfer rates.

(4) 'X' introduces filtered gas beneath the impeller so that bubbles are sheared and dispersed for maintaining anaerobic condition in every culture.

148. During a plasmid digestion analysis, a student made two sequential mistakes while performing gel electrophoresis.

Run 1: The student accidently poured only buffer (without agarose) into the casting tray, loaded the samples, and performed electrophoresis.

Run 2: After realizing the error made in run 1, the student prepared a proper agarose gel and repeated the electrophoresis under identical conditions, but this time, the electrodes were connected in reverse.

Given the circumstances, select the correct set of observations that are most consistent with these errors.

	Run 1	Run 2
(1)	Sample will remain inside the wells	Bands will appear near the wells
(2)	Faint bands will be visible under UV light	Bands will appear brighter due to reversed polarity
(3)	No bands will be visible under UV light	No bands will be seen in the gel lane
(4)	Samples will form a smear near the bottom of the tray	Bands will remain within gel lane but appear in inverted size order

(1) (1)

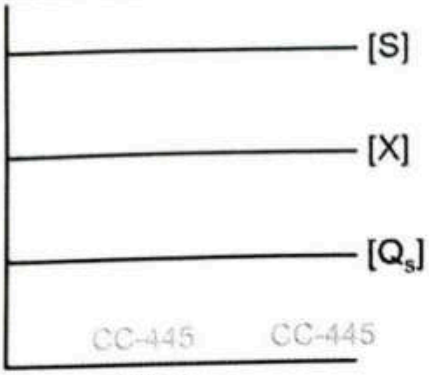
(2) (2)

(3) (3)

(4) (4)

Vivek Krishna
CC-445

149. The graph given below shows the cell biomass [X] and rate of substrate concentration [S] and rate of substrate consumption [Q_S] in a continuous culture system performed in a bioreactor.



Analyse it and choose the **incorrect** option.

(1) For performing this type of culturing, the design of the bioreactor is made such that the fresh lot of nutrient media is added from one side and the used media is removed from the other end to ensure a continuous supply of the desired product.

(2) During steady state, rate of growth and product formation, substrate concentration and rate of substrate utilisation remain constant.

(3) In this mode of culturing, the cultured cells are maintained in their physiologically most active log/exponential phase.

(4) This type of culturing method produces a smaller biomass, leading to low yield of desired protein.

150. Choose the **incorrect** match.

(1)	β-galactosidase	–	A reporter enzyme
(2)	Cloning	–	Production of phenotypically identical individuals with different genotypes
(3)	Heterologous host	–	An organism that is genetically engineered to express a gene or protein that is not native to it
(4)	DNA isolation	–	Purified DNA ultimately precipitates out after the addition of chilled CH_3CH_2OH that can be removed by spooling

(1) (1)

(2) (2)

(3) (3)

(4) (4)

151. All of the following steps are a part of downstream processing during the large-scale production of bio-pharmaceuticals, **except**

(1) Purification of product *via* chromatography

(2) Separation of product *via* centrifugation

(3) Formulation of product

(4) Expression of product

152. A cloning vector derived from pBR322 is modified in two ways: **Variant P** retains the native *ori*, while **Variant Q** has modified *ori* that increases the copy number by 5-10 fold. After transformation of these variants in *E.coli* and plating on ampicillin containing medium, the correct outcome observed will be that the

(1) Both variants will replicate equally and form same number of colonies as *ori* regulates replication

(2) Variant 'P' will produce higher of yield protein because *ori* majorly controls the rate of transcription

(3) Variant 'Q' will yield higher recombinant protein per cell because *ori* governs initiation of replication and copy number

(4) Both variants will replicate independently of the *ori* sequence, producing the same number of colonies

CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445 CC-445
Vivek Krishna

153. *Agrobacterium tumefaciens* is a rod-shaped gram-negative soil bacterium that is not/does not

(1) Capable of naturally transferring T-DNA into the genome of the host organism

(2) Contain a T_i-plasmid

(3) A pathogen of several plants like wheat

(4) Transform normal plant cells into a tumor

154. Muscular tissue
(a) Produces body movements
(b) Maintains posture
(c) Generates heat
(d) Primarily involve in locomotory actions
How many of the above is/are **correct**?

(1) One

(2) Three

(3) Two

(4) Four

155. In the human neural tissue, the neuroglial cells

(1) Are the unit of nervous system

(2) Make up less than 1/4th the volume of the neural tissue

(3) Receive electrical disturbances generated *via* the output zone and transmit the stimuli further

(4) Protect, nourish, insulate and support neurons

156. _____ junctions mainly help epithelial surfaces to resist separation during various contractile activities, such as when food moves through the intestine.
Select the **correct** option to fill in the blank.

(1) Synaptic

(2) Gap

(3) Adhering

(4) Communication

157. An epithelium

(1) Always has a free surface

(2) Does not provide protection, but perform only secretory activity

(3) Is a vascularized structure with minimal intercellular matrix

(4) Facilitates transcellular transport but does not have polarity

158. Consider the following statements
(a) Mammary gland is a unicellular gland.
(b) All the cartilages in vertebrate embryos are replaced by bones in adults.
(c) All involuntary muscle fibres are uninucleated, but all uninucleated muscle fibres are not branched.
(d) All connective tissues have fibres that provide strength, elasticity and flexibility to the tissues.
Select the best suitable option.

(1) (a) is false as mammary gland consists of isolated glandular cells.

(2) (b) is false as hyaline cartilage connects the floating ribs to the sternum.

(3) (c) is true as cardiac and visceral muscle fibres are involuntary and uninucleated in which smooth muscle fibres are non-branched.

(4) (d) is true as all connective tissues contain collagen and elastin.

159. Bone is classified under the category of specialised connective tissue. It differs from cartilage as

(1) Latter is characterized by the presence of lamellae

(2) Only former contains lacunae

(3) Former is the main tissue that provides structural frame to the body

(4) Only latter has calcium rich matrix

160. 'X' is the type of dense connective tissue that attaches thigh muscles to bones whereas 'Y' is the type of dense connective tissue that attaches bone to bone.
Identify 'X' and 'Y' and choose the **incorrect** option w.r.t. them.

(1) 'X' stabilizes joints and limits excessive motion

(2) 'Y' has higher elastin content as compared to 'X'.

(3) Both 'X' and 'Y' have fibroblasts that appear in rows between the bundles of collagen fibres

(4) 'Y' can get damaged by excess pressure and lead to injury like sprain

161. Match column I with column II w.r.t. types of epithelial tissues and their respective location in the human body.

	Column I		Column II
a.	Simple squamous epithelium	(i)	Lining of PCT
b.	Simple cuboidal epithelium	(ii)	Inner lining of intestine
c.	Simple columnar epithelium	(iii)	Alveoli of lungs
d.	Ciliated columnar epithelium	(iv)	Lining of oviduct

Select the **correct** option.

(1) a(iii), b(iv), c(i), d(ii)

(2) a(ii), b(iv), c(i), d(iii)

(3) a(i), b(iv), c(ii), d(iii)

(4) a(iii), b(i), c(ii), d(iv)

Vivek Krishna

162. Read the following facts
(a) Form the subcutaneous layer
(b) Have cells and fibres compactly arranged in a fluid ground substance
(c) Contain adipocytes, plasma cells and macrophages
(d) Undergo atrophy in an obese person
How many of the above mentioned facts is/are true and common between areolar and adipose tissues?

(1) Zero

(2) Three

(3) Two

(4) Four

163. The type of epithelial tissue that is present in the inner layer in wall of our blood vessels is characterized by

(1) Spindle-shaped nucleus

(2) Tall and slender cells

(3) Polygonal cells with irregular boundaries

(4) Basal nuclei and tessellated appearance

164. The Green Revolution succeeded in increasing the food supply, but it was not enough to feed the growing human population. Also,

(1) During this period, increased yields were mainly due to the use of improved crop varieties

(2) Use of agrochemicals was promoted during this period

(3) In India, this revolution was launched under the guidance of M. S. Swaminathan who introduced Russian variety of wheat

(4) Use of better management practices had a very insignificant role to play in increasing food supply during this period

165. Large scale production of genetically identical plants in a short time through tissue culture is possible because

(1) Somaclonal variation increases genetic diversity

(2) All regenerated plants arise from meiotically dividing cells of a single parent

(3) Plant cells are totipotent and in controlled nutritional and environmental conditions can grow and multiply continuously

(4) Tissue culture is a form of sexual reproduction that leads to the birth of a new generation

166. **Assertion (A)**: To obtain virus-free plants from a diseased variety, the apical meristem should be used as the explant in tissue culture.
Reason (R): Most plant viruses seldom invade the apical meristem because it lacks functional vascular connections and its slowly dividing cells restrict viral replication.
In the light of above statements, select the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

167. Genetically modified (GM) crops contribute to sustainable agriculture primarily because

(1) They increase yield only by high fertilizer usage

(2) They enhance resistance to pests and abiotic stresses, reducing dependence on chemicals

(3) They reduce soil fertility by converting nitrogen directly from the atmosphere

(4) They eliminate the need for irrigation and pollination entirely

168. Complete the analogy by selecting the correct option w.r.t the orders of the insects on which the Bt toxins work.
Dipterans : Mosquitoes :: Coleopterans : _____.

(1) Tobacco budworms

(2) Flies

(3) Armyworms

(4) Beetles

169. During screening of bacterial colonies carrying recombinant plasmids, a radiolabelled DNA probe complementary to the target gene was used. After complete hybridization and autoradiography, only a few colonies produced dark spots on the X-ray film. This observation can be explained by the fact that

(1) Only those colonies contain plasmids that successfully took up the foreign DNA complementary to the probe

(2) All colonies contain recombinant plasmids, but only some express the gene

(3) The probe binds non-specifically to bacterial genomic DNA, giving random dark spots

(4) Colonies showing dark spots have higher growth rate and therefore produce more copies of plasmid

170. There are an estimated _____ varieties of rice in India alone and _____ documented varieties of Basmati are grown in India.
Select the **correct** option to fill the respective blanks.

(1) 20 thousand; 27

(2) 0.2 million; 27

(3) 2 billion; 27

(4) 20 crore; 27

Vivek Krishna
CC-445

171. A transgenic tobacco plant was engineered to express dsRNA complementary to an essential mRNA of *Meloidogyne incognita*. When this plant is grown in *Meloidogyne* infested soil, it remained healthy. This demonstrates that the dsRNA

(1) Activates post-transcriptional silencing of mRNA of the plant

(2) Silences expression of a vital nematode gene after ingestion, preventing its survival

(3) Mutates nematode DNA, causing lethal gene loss

(4) Induces post-translation silencing of a vital gene of nematode

172. The transgenic cow, Rosie, produced milk that was enriched with what amount of human-protein?

(1) 24 gm/mL

(2) 0.0000024 kg/mL

(3) 2.4 kg/L

(4) 0.00024 kg/mL

173. Select the **incorrect** option.

(1) Mature insulin – 2 interchain and 2 intrachain disulphide bonds present

(2) Meristem culturing – Used for the production of banana, sugarcane and potato

(3) Protoplasts – Characterized by the presence of plasma membrane but no cell wall

(4) Pomato – Produced by somatic hybridization

174. In Bt cotton, the *cry* genes from *Bacillus thuringiensis* are introduced into the plant genome. Now, the resistance to insect pests results as

(1) The plant produces Bt spores that infect the insect's hindgut

(2) The plant synthesizes an inactive toxin that gets activated in the insect's gut, damaging its midgut epithelial cells

(3) The plant releases volatile toxins that repel insects before feeding

(4) The plant alters its surface wax to prevent insect attachment

175. ELISA is a technique that is used for disease detection. This technique

(1) Cannot measure the circulating hormone levels

(2) Works on the principle of antigen-antibody interaction

(3) Cannot detect the pathogen of viral diseases

(4) Cannot detect tumor antigens

176. Consider the statements given below

Statement A: *B. thuringiensis* forms protein crystals throughout its growth phase.

Statement B: Most Bt toxins are insect-group specific, thus, the choice of genes depends upon the crop and the targeted pest.

Select the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Statement B is incorrect but statement A is correct

(4) Statement A is incorrect but statement B is correct

177. For ADA deficiency in adults, genetically modified lymphocytes need to be administered periodically because

(1) The inserted ADA gene gets permanently integrated into all body cells but is not expressed continuously

(2) Transformed lymphocytes have a limited lifespan, so the introduced gene is lost as the cells die.

(3) The viral vector used for administration fails to deliver the gene into the lymphocytes effectively.

(4) The immune system eliminates all genetically modified cells immediately after re-infusion

178. A scientist develops a transgenic plant with enhanced drought tolerance by inserting a novel gene sequence isolated from a desert bacterium. She applies for a biopatent for this. Which of the following correctly identifies the ethical and legal principles governing this case?

(1) Any naturally occurring gene sequence, if isolated and characterized, can be patented.

(2) The entire transgenic plant can never be patented, neither the modified inserted gene sequence.

(3) A patent can only be granted if the gene and the method are both entirely synthetic and not derived from any biological source.

(4) The transformation method and the application of the gene in crop improvement can be patented, but not the naturally occurring gene itself.

CC-445 CC-445 CC-445

179. A transgenic mouse carrying an additional copy of a human gene for a growth factor shows abnormal tissue overgrowth. This observation demonstrates that

(1) The inserted gene alters only the mouse's immune response to human proteins

(2) The human gene remains silent in non-human cells

(3) The inserted gene is functional and its expression in the mouse mimics the gene's role in humans, revealing its physiological effect

(4) The phenotype results due to the random insertion, and it does not necessarily indicates the gene's function

180. During the production of recombinant human insulin, A-chain and B-chain were produced separately in *E. coli* and later combined by disulfide linkages.
Why was it necessary to produce them separately rather than expressing the gene directly?

(1) *E. coli* cannot process the inactive precursors to form mature insulin

(2) The 'A' and 'B' chains can form disulfide bonds inside the bacterial cytoplasm

(3) The original gene sequence becomes highly unstable inside *E. coli* plasmids and continously expresses itself

(4) Separate synthesis of 'A' and 'B' chains increases the overall yield of insulin per cell

CC-445 CC-445 CC-445 CC-445 CC-445 CC-445
Vivek Krishna